London Business School

Copying the US Senate

Backtest of senators' disclosed stock trades, September 2021 to August 2026

Study group B12
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1 Executive summary

US senators are required to disclose their securities transactions within 45 days. This report assesses whether an investor can profit from these disclosures by replicating them: buying a stock once a senator's purchase is made public and selling it once the same senator's sale is made public (the "Strategy"). The positions held in this way form the "Copy Portfolio". Our analysis covers disclosures made between 1 September 2021 and 31 August 2026 (the "Window").

As the capital invested in the Copy Portfolio varies over time, its performance is measured in two ways. Series A, the headline measure, is the return per dollar actually invested, comparable to that of a fund always fully invested in the stocks senators currently hold. Series B is the return of a fixed account, sized to fund every purchase, whose uninvested cash earns the Treasury bill rate (Section 2.4). The main benchmark is SPY, the S&P 500 ETF, both as held and risk-matched, i.e. scaled to the same volatility as the Copy Portfolio, so that both alternatives carry the same level of risk. The rules were designed on the first half of the Window (the "Design Period") and fixed before the second half (the "Test Period") was examined, so that the Test Period provides an independent check. Table 1 summarises the main results.

Table 1: Key figures, September 2021 to August 2026

Measure	Value	Measure	Value
Series A return, p.a.	13.4%	Series A CAPM alpha, p.a. (t-statistic)	+1.1% (0.18)
SPY return, p.a.	12.7%	Series A four-factor alpha, p.a. (t-statistic)	+2.3% (0.37)
Risk-matched SPY return, p.a.	16.3%	Series A market beta	1.21
Volatility p.a., Series A / SPY	23.4% / 15.9%	Test Period return, Series A / risk-matched SPY	22.9% / 26.0%
Sharpe ratio, Series A / SPY	0.50 / 0.61	Share of profit from two senators	77%

Note: Annualised geometric total returns in USD; alphas from monthly regressions, Newey-West t. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

We do not recommend investing in the Strategy, for the following reasons:

- The Copy Portfolio returned 13.4% p.a. against 12.7% for SPY, but with 1.5 times SPY's volatility. SPY held at the same level of risk returned 16.3% p.a., i.e. 2.9 percentage points p.a. more.
- The additional return over SPY reflects a market beta of 1.21. Once market exposure, and the size, value and momentum factors, are taken into account, the remaining return (alpha) is not statistically different from zero.
- The Test Period, which played no part in setting the rules, leads to the same conclusion: 22.9% p.a. for Series A against 26.0% for risk-matched SPY.
- Stocks bought by senators did not outperform the market between the transaction and its disclosure, which is where an informational advantage would be expected to appear first.
- Profit is concentrated: two senators generated 77% of it, and four semiconductor positions bought in early 2025 generated 26%. None of the 23 alternative specifications tested produces a statistically significant alpha.

2 Strategy definition

2.1 Regulatory context

Under the Stop Trading on Congressional Knowledge Act of 2012 (the "STOCK Act"), senators must report any securities transaction above USD 1,000 no later than 45 days after its execution, through a Periodic Transaction Report ("PTR") filed on the Senate's Electronic Financial Disclosures system ("eFD"). Each PTR states the security, the transaction date, the type of transaction (purchase, full sale or partial sale) and its value within a predefined band (USD 1,001 to 15,000, USD 15,001 to 50,000, etc.). A PTR becomes public when

eFD receives it, and eFD records the date and time of receipt. This is the earliest point at which an investor can act on the information, and the reference date used throughout this report.

2.2 Sample period and research design

The analysis covers PTRs made public between 1 September 2021 and 31 August 2026 (60 months). Five years provide 60 monthly observations for the regressions in Section 5 while remaining representative of the post-STOCK Act environment; the end date is the last month with factor data available.

Starting with an empty portfolio would leave only a handful of positions in the first months and would ignore sales of stocks bought earlier. The portfolio is therefore built from the PTRs made public in the 12 months preceding the Window (September 2020 to August 2021). Performance is measured only from 1 September 2021, by which date the portfolio held 70 positions. We refer to these 12 months as the warm-up period.

A strategy designed and evaluated on the same data tends to overstate its performance, as the rules may unintentionally fit that data. We therefore split the Window at its midpoint. All rules were designed using the first 30 months, September 2021 to February 2024 (the "Design Period"). The last 30 months, March 2024 to August 2026 (the "Test Period"), were kept out of the analysis and only examined once the rules had been frozen on 6 October 2026. Results over the Test Period are therefore out of sample. The chronology of decisions is set out in Appendix A.

2.3 Trading rules

The Strategy is fully mechanical. Table 2 sets out each rule and the reason it was adopted. Appendix B traces one copied transaction from disclosure to sale.

Table 2: Trading rules and rationale

Rule	Specification	Rationale
Signal	Every electronic PTR, dated by its time of receipt on eFD. The 308 PTRs filed on paper (scanned images) are excluded.	Receipt is when the information becomes public. Scanned PTRs cannot currently be parsed automatically, so an investor running the copy-trade strategy would be unable to replicate these trades through an automated process.
Scope	Transactions by individuals serving as senator on the transaction date, based on official Senate service dates.	eFD labels filers by their current status. Relying on it would exclude senators who have since left office, a survivorship bias.
Accounts	The senator's own and joint accounts.	These transactions are decided directly by the senator. Including spouse and child accounts is tested as an alternative (Section 6.5).
Instruments	US-listed common stocks, ADRs and equity ETFs. Options, bonds, preferred shares, mutual funds, crypto-assets and bond or commodity ETFs are excluded.	The Strategy is an equity strategy. Options and bonds cannot be sized or priced reliably from a value band.
Entry	Adjusted closing price on the first NYSE trading day after the day of receipt.	53% of PTRs are received after 15:30 New York time, when that day's close can no longer be traded. Using the next close avoids look-ahead bias. Same-day entry for earlier filings is tested as an alternative.
Position size	Midpoint of the disclosed band (USD 8,000.50 for the lowest band); lower bound for the open-ended top band.	Only the band is disclosed, and its midpoint is the central estimate. The lower bound and an equal amount per trade are tested as alternatives.
Exit	Sale when the same senator discloses a sale of the same stock, matched first in, first out. A full sale closes the position; a partial sale reduces it by the band midpoint, capped at the position held.	The Copy Portfolio follows the senator's own holding period, the only exit signal the filings provide.
No short sales	A disclosed sale of a stock not held in the Copy Portfolio is ignored.	Senators sell mainly for tax, liquidity or diversification reasons rather than on a negative view. After disclosure, sold stocks moved in line with SPY (+0.15% relative to SPY over 21 trading days, $t = 0.27$), so a short position would have earned nothing while incurring borrowing costs.
Departure from office	Positions remain open until the senator discloses a sale or the Window ends.	Former senators no longer file PTRs, so no sale signal is received. This case was not anticipated when the rules were set; its effect is quantified in Section 6.4.
Amendments	Each record takes effect as first published. A later correction reverses the original transaction on its own date and applies the corrected record.	An investor can only act on what has been published. Applying corrections retroactively would introduce look-ahead bias.

Note: Share of PTRs received after 15:30 and returns after disclosed sales computed over the Window. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

2.4 Return measurement

The capital invested in the Strategy is not fixed: positions are opened as purchases are disclosed and closed as sales are disclosed. Capital in positions was USD 1.5m on 1 September 2021 (the market value of the 70 positions built during the warm-up period) and rose to USD 24.9m by August 2026 (Appendix C). A return therefore requires a choice of denominator, and we report two measures, both computed on exactly the same transactions.

The headline measure, Series A, is the return on capital invested. Each day, Series A is the dollar-weighted return of the positions held since the previous close. For example, a USD 10,000 position rising by 2% and a USD 30,000 position falling by 1% give a return of −0.25% for the day. An investor can earn this return with a fixed amount of capital by holding each copied position in proportion to its disclosed size: when a new purchase is disclosed, the existing positions are scaled down pro rata to fund it, and when a sale is disclosed, the proceeds are reinvested pro rata in the remaining positions. The portfolio is therefore always fully invested and needs no subscriptions or withdrawals; the rebalancing this requires is included in the turnover reported in Section 5.4. Series A measures the quality of the signal per dollar invested and is directly comparable with fully invested benchmarks such as SPY, which is why the recommendation rests on it. A day without any position would earn the risk-free rate; owing to the warm-up period, this never occurs in the Window.

The supplementary measure, Series B, is the return of a fixed account of USD 17.1m: the USD 1.5m of initial positions plus USD 15.6m of cash, the largest cumulative net amount the Strategy ever needed to fund its purchases. Purchases are paid from the cash, sale proceeds return to it, and the cash earns the one-month Treasury bill rate, credited daily. Series B reflects a funded investor's experience, but its size could only be known with hindsight and 91% of it was in cash at the start, so its early performance reflects the cash balance rather than the trades. It is reported as a supplementary measure only.

2.5 Benchmarks

Each benchmark answers a different question, and all are measured over identical dates. SPY, the SPDR S&P 500 ETF, shows whether a passive investor in the US market would have done as well. Risk-matched SPY is the risk-free rate plus k times SPY's return above the risk-free rate, with k set so that its volatility equals that of the series being assessed (k = 1.48 for Series A and 0.51 for Series B); it shows whether a higher return is simply compensation for higher risk, and is the main basis of our recommendation. The Senate universe holds every stock bought by any senator in the previous 12 months, equally weighted and rebalanced monthly using only filings already public; it shows whether the timing of senators' trades adds anything beyond the list of stocks they buy.

We also regress the monthly return of Series A above the risk-free rate on the market factor (CAPM) and on the market, size (SMB), value (HML) and momentum (MOM) factors (Fama and French, 1993; Carhart, 1997). The intercept, alpha, is the part of the return not explained by exposures that an investor could buy cheaply through index and style funds, so this model serves as our style-matched benchmark. Factor returns and the risk-free rate (the one-month US Treasury bill) come from the US files of the Kenneth French Data Library, as in the course. Finally, NANC and KRUZ, two ETFs launched in February 2023 that copy the disclosed trades of Democratic and Republican members of Congress respectively, show what a professional implementation of the same idea achieved.

3 Economic rationale and testable predictions

The Strategy can only work if senators trade on information that is not yet reflected in prices and is still not reflected when their filing is published. Senators sit on committees that draft regulation, award public contracts and question company executives. Ziobrowski et al. (2004) estimated that senators' purchases outperformed the market by approximately 12% p.a. between 1993 and 1998, while studies covering the period after the STOCK Act find little or no outperformance (Eggers and Hainmueller, 2013; Belmont et al., 2022).

The natural question is who is on the other side of the trade. A copier buys about one month after the senator, from whoever sells on that day. For these sellers to lose consistently, two conditions must hold: the senator's information must still be unpriced at the time of disclosure, and other investors must ignore a public filing that is tracked by retail applications, newsletters and two listed ETFs. The first condition is plausible; the second is difficult to sustain.

Each possible explanation implies a different pattern in the data. On 30 September 2026, before the Test Period was examined, we recorded five hypotheses and the result each explanation predicts. If senators are informed and the copier acts in time, (i) the Copy Portfolio earns a positive alpha (H0), (ii) stocks keep outperforming after disclosure (H2), (iii) sales in industries overseen by the senator's committees underperform other sales (H3), and (iv) larger purchases, which signal greater conviction, outperform smaller ones (H4). If senators are informed but prices adjust before the filing, purchases outperform between the transaction and its disclosure (H1), and only the senator benefits. If the return reflects market exposure only, alpha is zero and the factor loadings explain the return. Section 6.2 compares each prediction with the result.

4 Data and sample construction

All 1,876 PTRs received by eFD between January 2016 and September 2026 were collected directly from the Senate's website: 1,568 filed electronically and 308 filed on paper. After removing duplicates created by amendments and applying the rules in Table 2, the Window contains 2,340 transactions by 31 senators (Appendix D). Fifteen of these senators disclosed only sales of stocks not held in the Copy Portfolio and therefore generate no positions. The 85 paper PTRs filed within the Window, 64 of them by Richard Blumenthal, cannot be replicated systematically (as mentioned in 2.3).

The sample is highly concentrated. Tommy Tuberville, Markwayne Mullin and Alan Armstrong account for 87% of the dollars invested (Appendix E). Alan Armstrong was appointed on 24 March 2026 to the seat vacated by Markwayne Mullin. A single PTR he filed on 21 July 2026 reports 654 purchases of 621 stocks made in late March, as he established his portfolio on taking office. This filing represents 67% of the purchases in the Test Period, so every transaction-level result is also reported without it.

Prices are daily adjusted closing prices from Yahoo Finance, which include reinvested dividends; returns are therefore total returns in US dollars. For a sterling-based investor the comparison is unchanged (Series A 13.8% p.a., SPY 13.1% p.a.). Adjusted prices reflect corporate actions as revised to September 2026, while all filings are used as first published. One risk-free rate, the one-month US Treasury bill from the Kenneth French Data Library, is used throughout. The portfolio is valued daily, while Sharpe ratios and regressions use monthly returns, consistent with holding periods of several months; volatility is annualised by $\sqrt{12}$, and t-statistics are adjusted for autocorrelation (Newey and West, 1987).

Yahoo Finance holds no price history for delisted companies, so purchases of stocks that were later acquired or delisted could not be valued from that source. At their peak, in October 2022, such stocks would have represented 67% of capital invested. Daily total-return data from Bloomberg were therefore used for the eight largest gaps, which cover 95% of the missing dollars; the 178 smaller stocks that remain unpriced never exceed 2.2% of capital. Seven of the eight were acquired, and the positions in U.S. Steel, ChannelAdvisor and Activision Blizzard rose on the takeover bids. Including them raises Series A from 10.8% to 13.4% p.a.: the missing data biased the return downwards, not upwards. As this correction was made after the Test Period had been opened, all values before and after it are reported in Appendix F. No conclusion changes.

5 Results

5.1 Performance against benchmarks

This section establishes whether the Copy Portfolio earned more than an investor could have obtained with the same money at the same risk. Figure 1 shows the growth of USD 1 invested in Series A and in each benchmark, and Table 3 reports the performance statistics over identical dates.



Figure 1: Growth of USD 1 invested on 31 August 2021, daily total return

Note: Risk-matched SPY is scaled to the volatility of Series A (k = 1.48). The dashed vertical line separates the Design Period from the Test Period. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

Table 3: Performance against benchmarks, September 2021 to August 2026 (60 months)

	Series A (headline)	Series B (fixed account)	SPY	SPY risk-matched to Series A (k = 1.48)	SPY risk-matched to Series B (k = 0.51)	Senate universe
Return and risk						
Return, % p.a.	**13.41**	**9.16**	**12.69**	**16.30**	**8.46**	**13.56**
Volatility, % p.a.	23.41	8.03	15.87	23.47	8.09	17.14
Sharpe ratio	**0.50**	**0.69**	**0.61**	**0.61**	**0.61**	**0.62**
Factor models (monthly regressions, Newey-West t-statistics)						
CAPM alpha, % p.a. (t)	+1.07 (0.18)	+2.28 (1.08)	+1.02 (1.62)	+1.52 (1.62)	+0.52 (1.62)	+1.72 (0.74)
CAPM beta	1.21	0.37	0.98	1.45	0.49	1.01
Four-factor alpha, % p.a. (t)	+2.26 (0.37)	+2.62 (1.25)	+0.09 (0.21)	+0.14 (0.21)	+0.05 (0.21)	+1.68 (0.77)
Size (SMB) loading (t)	0.33 (1.87)	0.18 (2.33)	−0.13 (−8.29)	−0.19 (−8.29)	−0.06 (−8.29)	0.13 (1.87)
Smallest alpha detectable at t = 2, % p.a.	12.39	4.85	1.01	1.49	0.51	
Drawdown and diversification						
Maximum drawdown, %	−32.16	−7.13	−24.50	−35.31	−12.52	−24.58
Peak, trough, recovery	Mar 22, Jul 22, Mar 24	Jun 26, Jul 26, Aug 26	Jan 22, Oct 22, Dec 23	Jan 22, Oct 22, Jan 24	Jan 22, Oct 22, Jun 23	Nov 21, Oct 22, Dec 23
Correlation with SPY: all months / SPY's worst 12	0.81 / 0.68	0.73 / 0.36	1 / 1	1 / 1	1 / 1	0.94 / 0.87
Trading						
One-way turnover, % of capital p.a.	102	39				
Return over its risk-matched SPY, % p.a.	**−2.58**	**+0.64**				
Break-even one-way trading cost	none	82 bp				

Note: Annualised geometric returns; alphas are monthly estimates × 12. Return over risk-matched SPY is arithmetic (geometric: −2.9 pp for Series A); "none" means it is negative before costs. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

Over the Window, Series A outperformed SPY by 0.7 percentage points p.a. (13.4% against 12.7%), with a volatility of 23.4% against 15.9%. Adjusted for this difference in risk, SPY was 2.9 percentage points p.a. ahead (16.3% for SPY risk-matched to Series A), and the Sharpe ratio of Series A (0.50) is below that of SPY (0.61). The higher raw return is therefore a consequence of higher risk rather than of better selection.

The factor regressions identify the source of that risk. Series A has a market beta of 1.21, which explains the return above SPY, and a positive size loading of 0.33 that is suggestive of a tilt towards smaller companies but not statistically significant (t = 1.87). Both exposures are available at low cost through index and style funds. The CAPM and four-factor alphas (+1.1% and +2.3% p.a.) are each less than one standard error from zero. The equally weighted Senate universe earned a similar return (13.6% p.a.) with lower volatility, which

indicates that following the timing of senators' trades added little beyond holding the stocks they buy, although part of the difference may reflect the different weighting of the two portfolios. The power of these tests is limited: with 60 monthly observations, only an alpha of about 12% p.a. would have reached a t-statistic of 2. The results therefore rule out a large informational advantage but cannot rule out a small one (Section 6.5).

Series B shows a higher Sharpe ratio than SPY (0.69 against 0.61) and outperforms SPY risk-matched to its own volatility (9.2% against 8.5% p.a.). This reflects the composition of the account rather than the senators' trades: most of the account was in cash during the 2022 market decline, and as the portfolio grew the account became progressively more invested in a rising market. Its alpha (+2.3% p.a., t = 1.08) is not significant, its size could only be set with hindsight, and its advantage over risk-matched SPY turns negative in the Test Period (−1.8 percentage points p.a.).

5.2 Risk and contribution to an existing portfolio

An investor would typically hold the Strategy alongside an existing equity portfolio, so its drawdown and diversification benefit matter as much as its standalone return. The largest fall of Series A (32.2%, from March to July 2022) was deeper than that of SPY (24.5%), and Series A regained its previous peak in March 2024, three months after SPY (Appendix G). A small allocation improves an existing portfolio only if the asset's Sharpe ratio exceeds its correlation with the portfolio times the portfolio's Sharpe ratio: 0.503 against 0.81 × 0.61 = 0.499 here, a difference well within estimation error. In 10,000 bootstrap resamples of the 60 months, a 10% allocation to Series A improves the Sharpe ratio of an SPY holding in only 49% of cases, and the two Sharpe ratios are not significantly different (p = 0.69). The correlation with SPY falls only to 0.68 in SPY's twelve weakest months. Series A therefore offers neither a reliable improvement in risk-adjusted return nor a hedge.

5.3 Listed congressional-trading ETFs

NANC and KRUZ indicate what a professional implementation of a similar idea achieved. Over the 42 months since their launch, NANC returned 24.8% p.a. (Sharpe ratio 1.28) and KRUZ 19.3% (1.05), against 22.3% (1.33) for SPY and 19.3% (0.79) for Series A over the same months. Neither ETF has an alpha statistically different from zero (CAPM alpha +0.84% p.a., t = 0.44, and −1.33% p.a., t = −0.39). Both track the market closely: their monthly correlations with SPY are 0.96 and 0.89, and their correlations with Series A are 0.77 and 0.88. Professional implementations have therefore also delivered market returns rather than an excess return.

5.4 Implementation costs

Trading costs would reduce the return further. Series A turns over approximately 102% of its capital per year, one way, including the pro-rata rebalancing needed to keep it fully invested. No return net of costs is reported, as any precise cost assumption would not be supported by observable data, and none is needed: since Series A already underperforms risk-matched SPY before costs, there is no positive cost at which it would break even. Series B breaks even at 82 basis points per trade over the Window, but not in the Test Period. Costs therefore reinforce the recommendation.

6 Robustness and out-of-sample evidence

6.1 Test Period and year-by-year performance

A result that holds only in the data used to design the rules, or only in a few favourable years, would not support an investment. Table 4 reports the two halves of the Window and each calendar year.

Table 4: Performance of Series A by period and by calendar year

Period	Series A	SPY	SPY risk-matched to Series A	Series A CAPM alpha (t)
By half of the Window (annualised)				
Design Period, Sep 2021 to Feb 2024	4.61%	6.43%	6.85%	+1.14% (0.12)
Test Period, Mar 2024 to Aug 2026	22.94%	19.32%	26.02%	+1.69% (0.29)
By calendar year (not annualised)				
2021 (Sep to Dec)	−2.25%	+5.89%	+8.51%	
2022	−6.01%	−18.18%	−27.63%	
2023	+19.94%	+26.18%	+36.85%	
2024	+5.02%	+24.89%	+35.02%	
2025	+27.88%	+17.72%	+24.29%	
2026 (Jan to Aug)	+26.76%	+13.08%	+17.99%	
Worst 12 months of Series A (Nov 2021 to Oct 2022)	−13.73%	−14.63%		

Note: Risk-matched SPY uses Series A's volatility in each period (Test Period k = 1.44). Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

The Test Period confirms the overall result. Series A outperformed SPY by 3.6 percentage points p.a. but underperformed SPY at the same volatility (22.9% against 26.0% p.a.), and its Sharpe ratio is below SPY's in both halves (0.20 against 0.27, then 1.03 against 1.17). The four-factor model attributes the Test Period return to market exposure and to momentum (loading 0.38, t = 3.47), i.e. to buying stocks that had recently risen. Year by year, performance is unstable: Series A trailed SPY in 2021, 2023 and 2024 (by 19.9 percentage points in 2024), lost less than SPY over calendar 2022 despite a deeper peak-to-trough fall within the year, and outperformed in 2025 and 2026, which carry most of the five-year result.

6.2 Pre-registered hypothesis tests

The tests below determine which of the explanations in Section 3 the data support. Table 5 sets each of the seven statistics recorded before the Test Period was opened against the prediction and against the outcome expected if senators traded on information still available to a copier.

Table 5: Registered predictions against Test Period results (March 2024 to August 2026)

Test	Expected if senators informed	Registered prediction	Result (t)	Excluding bulk filings (t)	Adjusted p-value	Consistent with prediction
H0 CAPM alpha	Positive	Zero	+1.69% p.a. (0.29)		1.000	Yes
H0 Four-factor alpha	Positive	Zero	+0.40% p.a. (0.06)		1.000	Yes
H1 Purchases, transaction to disclosure	Positive	Close to zero	−3.22% (−2.98)	−0.11% (−0.18)	0.017	No (opposite sign)
H1 Sales, transaction to disclosure	Negative	Close to zero	−0.99% (−0.96)	+0.68% (1.22)	1.000	Yes
H2 Purchases, 21 days after disclosure	Positive	Sign tested	+1.74% (3.54)	+1.12% (0.81)	0.001	Partly
H3 Committee sales, overseen less other	Negative	No difference	+6.16% (0.94)		1.000	Yes
H4 Small less large purchases, 126 days	Negative	Negative only if informed	+4.82% (0.96)	+1.72% (0.41)	1.000	Yes

Note: Returns relative to SPY, t clustered by month (H0: Newey-West; H2 one-sided). p-values adjusted for seven tests (Holm, 1979). Bulk filings list 50+ transactions. See Appendix H. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

Five of the seven statistics match the predictions. Had senators traded on information a copier could still use, several of these results would point the same way; none does. The only significant departure from the predictions, H1 for purchases, has the opposite sign: stocks bought by senators underperformed the market before disclosure. H2 is positive, but both significant results stem from Armstrong's filing, whose 621 stocks fell relative to SPY between the March purchases and the July disclosure and then recovered. As two-thirds of the events fall in a single month, standard errors clustered by month overstate precision. Excluding bulk filings,

H1 is −0.11% and H2 +1.12%, neither statistically significant. The evidence is therefore most consistent with the explanation that the return reflects market exposure rather than information.

6.3 Senator returns and copier returns

The disclosure delay is often cited as the main cost of copying. Figure 2 measures it directly by comparing, for every purchase and sale, the stock's return relative to SPY from the senator's transaction date with its return from the copier's entry date.

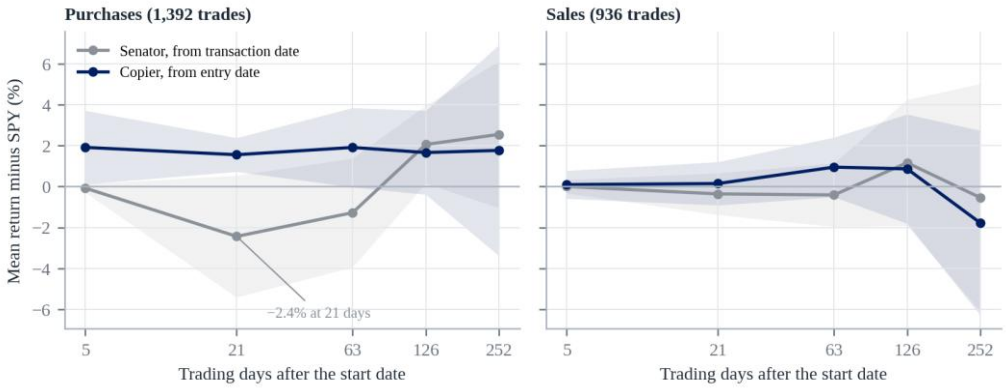


Figure 2: Return relative to SPY after the senator's transaction (grey) and after the copier's entry (navy), full Window

Note: Equal-weighted average across all transactions; shaded areas are 95% confidence intervals clustered by month of entry. Not a registered test. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

For purchases, waiting for the disclosure did not cost the copier anything; on average it helped. One month after the transaction, senators' purchases had underperformed SPY by 2.44%, while the same stocks bought at disclosure had outperformed SPY by 1.56% one month after entry. The difference of 4.01 percentage points (t = −2.25) arises because the stocks fell by 2.06% relative to SPY (t = −1.60) during the median 29 trading days between transaction and disclosure. The two lines converge after six and twelve months. The copier's early gains are equal-weighted, dominated by Armstrong's filing and not significant beyond three months, and they do not carry through to the dollar-weighted Series A. For sales, neither line moves materially, so senators do not appear to sell ahead of price falls. The pattern is the reverse of what informed trading would produce.

6.4 Contribution by senator and departures from office

Because the sample is concentrated, it matters which senators drive the result. Markwayne Mullin and Ashley Moody generated 77% of the profit earned in the Window (Appendix E). Four of Moody's 16 positions, in Micron, AMD and Applied Materials, bought within four months of her appointment in January 2025, generated USD 1.8m, or 26% of the total. Excluding all trades made in a senator's first 180 days in office lowers the return of Series A from 13.4% to 9.4% p.a. A large part of the profit therefore comes from new senators building their portfolios rather than from repeated, informed trading.

Mullin's positions illustrate how concentration affects the measured return. Removing them raises Series A to 18.6% p.a. although they made 39% of the profit. Series A measures the return per dollar invested. Mullin's positions returned 13% to 21% a year but made up about half of the capital in 2024 to 2026, while the rest of the portfolio, led by Moody's semiconductor holdings, gained 45% in 2025 and 45% in the first eight months of 2026. His large positions added dollars of profit but lowered the return per dollar (Table E2).

Departures from office add to this effect. Mullin left the Senate on 23 March 2026 and filed no PTR afterwards; under the rules, his 150 positions (USD 6.0m at cost) stayed in the portfolio until the end of the Window. As this case was identified after the Test Period had been opened, its effect is reported as a supplementary check rather than as a change to the rules: selling a senator's positions at the first close after leaving office raises Series A to 16.0% p.a., with a CAPM alpha of +3.3% p.a. (t = 0.48). The Strategy then remains 1.3 percentage points p.a. below SPY at the same volatility (17.3%), and the recommendation is unchanged.

6.5 Alternative specifications and statistical power

A result that depends on one particular choice of rules is not reliable. We therefore changed one rule at a time and report every specification tested (Table 6; each is listed in Appendix I). Of the 23 specifications, 19 were defined before any return was computed, three after the Design Period results had been reviewed, and one, the

departure check in Section 6.4, after the Test Period had been opened. A further specification using CRSP prices could not be run, as they could not be retrieved programmatically under our WRDS access.

Table 6: Alternative specifications and their effect on Series A (full Window)

Category	Rule changed	Runs	Return, % p.a.	CAPM alpha, % p.a.	Largest |t|
Senators and filings	Excluding each of the five most active senators in turn; excluding senators' first 180 days in office; excluding bulk filings; including spouse and child accounts	8	9.4 to 18.6	−2.3 to +5.8	0.78
Instruments	Excluding ETFs; excluding stocks traded over the counter; including all ETFs, including bond and T-bill funds; including preferred shares	4	12.0 to 13.4	−0.1 to +1.1	0.18
Position sizing	Lower bound of the band; equal amount per trade; equal weight per position	3	12.4 to 13.4	+0.8 to +1.1	0.39
Timing	Same-day close for PTRs received before 15:30; sale when a senator leaves office	2	13.9 to 16.0	+1.5 to +3.3	0.48
Data and window	3-year and 10-year windows; amendments applied retroactively; unpriced stocks earn SPY's return; no warm-up period; Yahoo prices only	6	10.8 to 25.5	−1.5 to +3.0	0.52
Base case			**13.41**	**+1.07**	**0.18**

Note: Newey-West t-statistic of the CAPM alpha, as in all tables. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

No specification produces an alpha with $|t| \geq 2$, either over the Window or over its own Test Period. The largest is the four-factor alpha of the 3-year window over its 18-month second half (+14.9% p.a., t = 1.84), which rests on too few months to be informative.

Two further safeguards apply. First, only the seven registered statistics are treated as tests. The additional breakdowns we examined produced 516 statistics, of which 81 reach $|t| = 2$; these overlap heavily and are concentrated in groups dominated by Armstrong's filing, so they are not taken as evidence. Second, the sample is short: given the volatility of Series A not explained by the market, an alpha of 3% p.a. would require about 77 years of monthly data to reach t = 2, and an alpha of 5% p.a. about 28 years ($t \approx$ information ratio $\times \sqrt{\text{years}}$). The 24-month rolling alpha ranges from +10% to −16% p.a. and is significant only in early 2025, when it is negative (Appendix G). Five years of data cannot prove that senators have no informational advantage, but they rule out one large enough to justify an investment. Appendix J summarises the safeguards against common backtest biases.

7 Case against our recommendation and conditions for revisiting it

The strongest case for the Strategy is as follows. Series A outperformed SPY over the Window (13.4% against 12.7% p.a.) and over the Test Period (22.9% against 19.3% p.a.), lost less over calendar 2022, and has positive alpha estimates, which rise to 3.3% p.a. when positions are sold on a senator's departure. Series B has a higher Sharpe ratio than SPY. An investor who cannot borrow cannot hold 1.48 times SPY, and may regard the unadjusted comparison as the relevant one. Finally, the tests could not detect an alpha below about 12% p.a.

These arguments do not change our recommendation. First, the same level of risk can be obtained without borrowing, through high-beta or small-company funds, so the risk-matched comparison applies to an investor who does not borrow. Second, the Sharpe ratio of Series B reflects when the account happened to hold cash, largely during the 2022 decline, rather than the trades themselves. Third, senators' purchases did not outperform before disclosure, where informed trading would show first. Fourth, the profit depends on a few positions and on who sits in the Senate: one senator's departure changes the return by 2.6 percentage points p.a., and excluding a single senator by up to 5. Fifth, the two ETFs that implement the idea professionally show no alpha either.

Three pieces of evidence would justify revisiting the recommendation: (i) a positive return before disclosure on purchases by senators who file promptly, the clearest sign of information a copier could still capture (in our data +0.28%, t = 0.57, for trades filed within 30 days); (ii) prices from CRSP, which covers delisted stocks, for the 178 stocks still unpriced (see section 4); and (iii) a longer and broader sample, for example including the House of Representatives, large enough to detect an alpha of 3% to 5% p.a. that does not depend on the two largest contributors. Until then, an investor seeking this level of risk can hold about 1.5 times the index, which delivered the same volatility with 2.9 percentage points p.a. more return.

8 Limitations

Five limitations should be borne in mind. The sample covers only 60 months, 30 in each half, so only large alphas can be detected (Section 6.5). The rules include no exit when a senator leaves office; Section 6.4 shows that adding one raises the return by 2.6 percentage points p.a. without changing the recommendation. 178 stocks remain unpriced, at most 2.2% of capital, and since pricing the largest gaps raised the return, any remaining bias most likely understates it. Armstrong's July 2026 filing dominates the transaction-level tests in the Test Period, although not the portfolio result (13.2% p.a. without him). Finally, Yahoo Finance revises its price history, so a single September 2026 download is used throughout; two renamed tickers (UTX and DWDP) include spin-offs that predate the Window and affect only the 10-year specification.

9 Conclusion

Between September 2021 and August 2026, copying senators' disclosed trades earned a return in line with the market, driven by a beta of 1.2, with higher volatility, a deeper drawdown and no statistically significant alpha in either half, in any year-by-year window or under any alternative specification. Senators' purchases did not outperform the market before disclosure, leaving nothing for a copier to capture, and the filings are available to every market participant. We therefore do not recommend investing in the Strategy. The question would merit further work with complete prices for delisted stocks and a sample long enough to detect a modest advantage. On the evidence available, the Strategy is a more volatile and more costly way of holding the index.

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Appendices

The appendices support the report and are not needed to follow it.

Abbreviations and defined terms

Term	Definition	Term	Definition
ADR	American Depositary Receipt	PTR	Periodic Transaction Report
CAPM	Capital Asset Pricing Model	p.a.	Per annum
Copy Portfolio	Positions opened by copying senators' disclosed purchases	Risk-matched SPY	SPY scaled to the volatility of the series it is compared with
CRSP	Center for Research in Security Prices database	Series A	Return per dollar invested (headline measure)
Design Period	September 2021 to February 2024	Series B	Return of a USD 17.1m fixed account
eFD	Senate Electronic Financial Disclosures system	SPY	SPDR S&P 500 ETF
ETF	Exchange-traded fund	STOCK Act	Stop Trading on Congressional Knowledge Act of 2012
HML / SMB / MOM	Value / size / momentum factors	Test Period	March 2024 to August 2026
NYSE	New York Stock Exchange	Warm-up period	September 2020 to August 2021, used to build the initial portfolio
OTC	Over the counter	Window	Disclosures from 1 September 2021 to 31 August 2026

A. Chronology of decisions

- 27 September 2026: Data source, scope, position sizing, entry and exit rules, benchmarks and the split of the Window fixed. The first 12 alternative specifications defined before any price or return had been computed.
- 28 to 30 September 2026: Four further alternative specifications defined before any portfolio return existed. The 12-month warm-up period and the exclusion of preferred shares decided after the Design Period results had been reviewed, and recorded as such.
- 30 September to 2 October 2026: The five hypotheses, their predictions and the adjustment for multiple testing recorded, before any Test Period result existed.
- 6 October 2026: Rules frozen and their exact version recorded; the Test Period was then opened.
- 9 October 2026: Bloomberg prices added for eight delisted stocks, with all values before and after the change recorded (Appendix F).
- 13 October 2026: Supplementary check of selling positions when a senator leaves office (Section 6.4); the rules are unchanged.

B. Worked example

Tommy Tuberville bought Johnson & Johnson shares (joint account, band USD 1,001 to 15,000) on 30 September 2021. The PTR was received at 13:32 on 15 October 2021, so the copier bought at the close of Monday 18 October at an adjusted price of USD 139.25, investing the USD 8,000.50 band midpoint in 57.45 shares. His full sale of 16 October 2023 was disclosed on 16 November 2023, and the copier sold at the close of 17 November at USD 137.55, a return of −1.22% over 525 trading days.

C. Capital invested

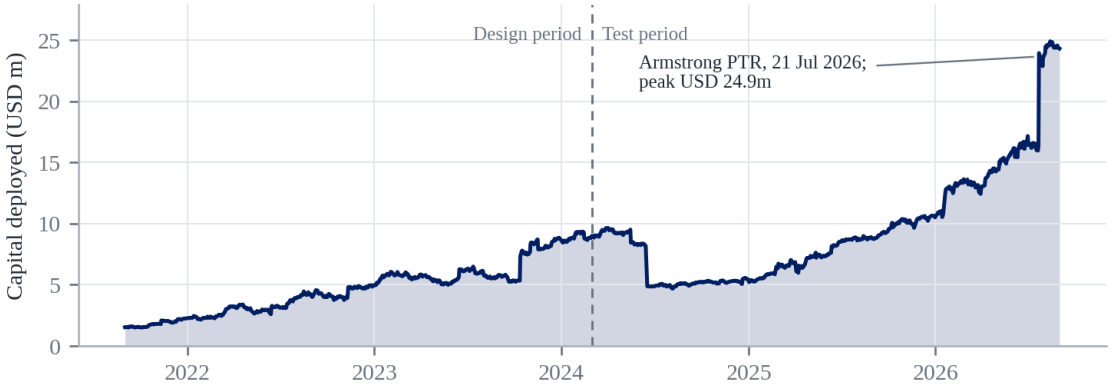


Figure C1: Capital invested in copied positions at each close, USD m

Note: The July 2026 increase changes the composition of Series A, not the scale of its return. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

D. Sample construction

Table D1: From all disclosed transactions to the sample used

Step	Removed	Remaining
All transactions in electronic PTRs received January 2016 to September 2026		14,592
Duplicates restated in amendments or filed twice	2,182	12,410
Transaction date outside the filer's Senate service	5	12,405
First disclosed outside 1 September 2021 to 31 August 2026	7,125	5,280
Asset other than stock (municipal bonds 693, options 326, corporate bonds 162, futures 141, other 34)	1,356	3,924
Options reported as stock, exchanges of shares, no identifiable ticker	292	3,632
Mutual funds, non-US listings, other instruments, preferred shares	71	3,561
ETFs other than equity funds (bond, T-bill, commodity, other)	187	3,374
Spouse or dependent-child accounts	**1,034**	**2,340**

Note: Of the 2,340 transactions, 31 have no price on the entry date and are not copied. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

E. Contribution by senator

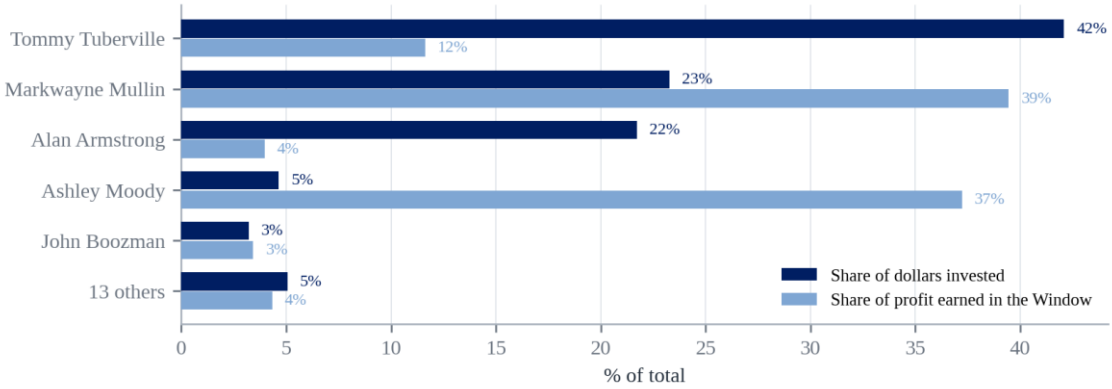


Figure E1: Share of dollars invested and of profit earned in the Window, by senator

Note: Dollars invested include warm-up positions; profit counts only gains and losses from 1 September 2021. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

Table E1: Number of transactions and disclosure lag by senator

	Transactions	Median lag (days)	Share filed after 45 days
All transactions in the sample	2,340	33	30.6%
Tommy Tuberville	732	31	0.0%
Alan Armstrong	702	113	99.9%
Markwayne Mullin	322	29	0.6%
John Boozman	251	25	1.6%
Sheldon Whitehouse	81	27	4.9%
All other senators	252	27	2.0%

Note: Lag is the disclosure date less the transaction date, in calendar days. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

Table E2: Series A return of Markwayne Mullin's positions and of all other positions, by calendar year

	2023	2024	2025	2026 (Jan to Aug)
Return of Mullin's positions	12.6%	20.7%	18.2%	14.4%
Return of all other positions	22.6%	−0.2%	44.6%	44.6%
Mullin's share of capital invested	9%	52%	59%	50%

Note: Series A recomputed for each group of positions; Mullin's first copied purchase was disclosed in 2023. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

F. Effect of Bloomberg prices for delisted stocks

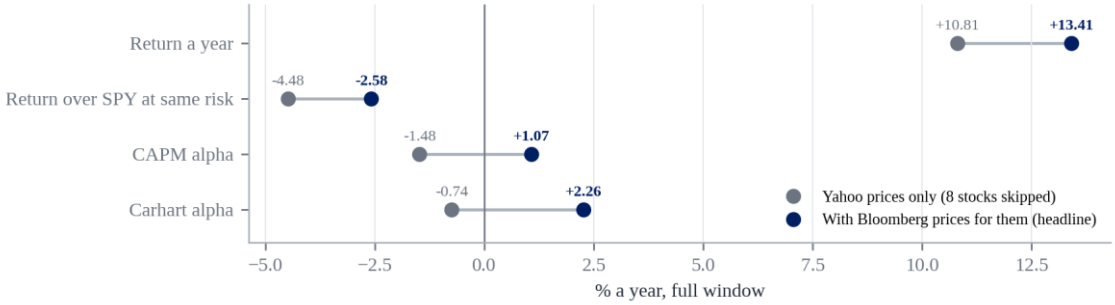


Figure F1: Results over the Window with Yahoo prices only and with Bloomberg prices for the eight delisted stocks, % p.a.

Note: Test Period return 22.95% before, 22.94% after; no result in Table 5 changes. Design Period Series A moves from −0.14% to 4.61% p.a.; predictions were not revised. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

G. Risk and stability

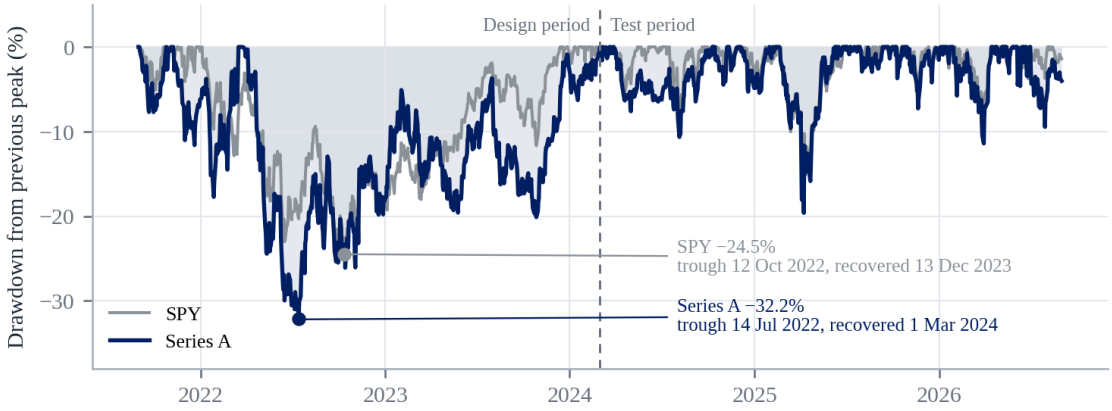


Figure G1: Decline from previous peak, Series A and SPY, daily

Note: Series A fell 32.2% to 14 July 2022 (recovered 1 March 2024); SPY fell 24.5% to 12 October 2022 (recovered 13 December 2023). Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

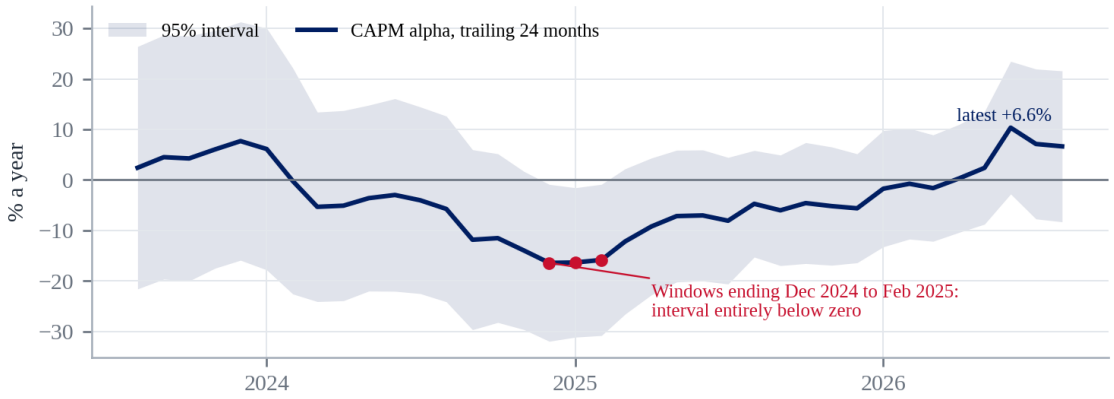


Figure G2: CAPM alpha of Series A over rolling 24-month windows, % p.a., with 95% confidence interval

Note: Each point is a regression on the previous 24 months; neighbouring points share 23 of 24 months. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

Table G1: Additional statistics on Series A

Statistic	Value
Sharpe ratio, Series A against SPY	0.50 against 0.61
Difference in Sharpe ratios, test of Jobson and Korkie (1981) with Memmel (2003) correction	−0.11 (p = 0.69)
Difference in Sharpe ratios, 95% bootstrap interval	−0.70 to 0.37
CAPM alpha, 95% bootstrap interval, % p.a.	−9.7 to 12.0
Change in SPY's Sharpe ratio from a 10% allocation to Series A	−0.0016; positive in 49% of resamples
Volatility of Series A not explained by the market, % p.a.	13.1
Years of data needed for t = 2 at an alpha of 2%, 3%, 5%, 10% p.a.	173, 77, 28, 7

Note: Monthly returns; stationary block bootstrap (Politis and Romano, 1994), 10,000 resamples. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

H. Registered tests with and without bulk filings

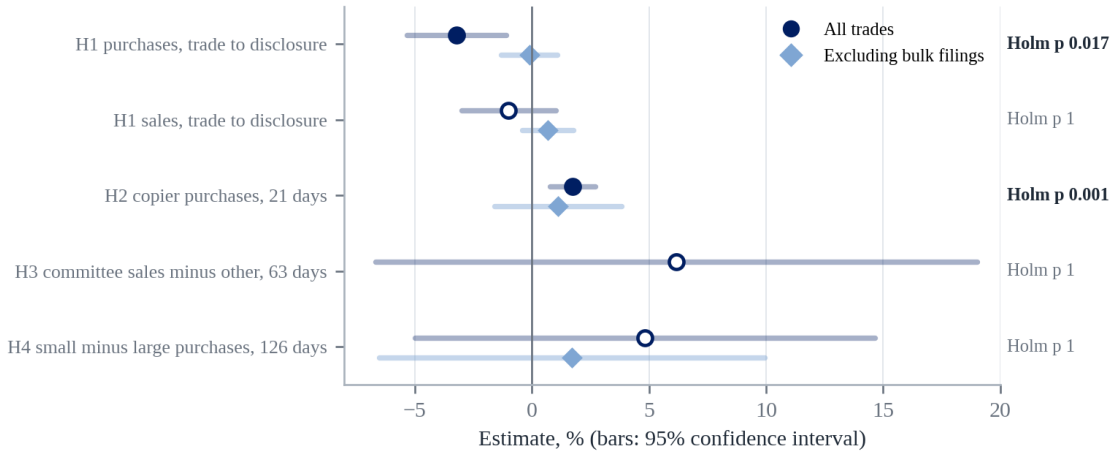


Figure H1: Test Period estimates with 95% confidence intervals, with and without bulk filings, %

Note: Filled circles are significant after adjustment for multiple testing. Bulk filings are PTRs listing 50 or more transactions. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

I. Alternative specifications

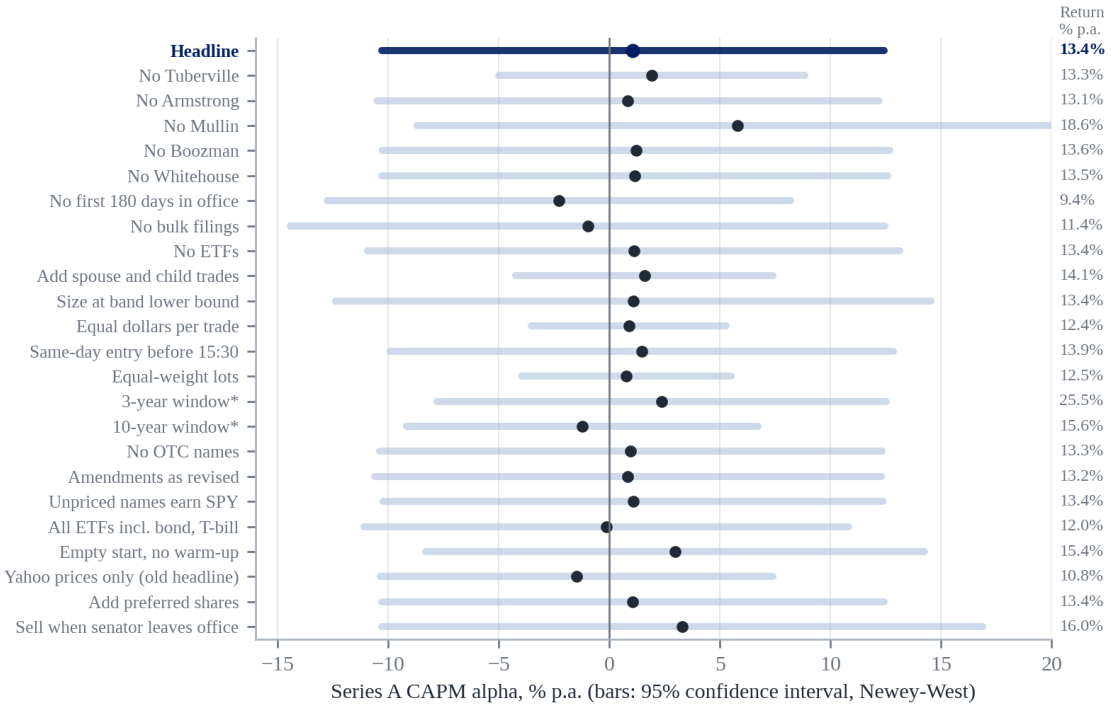


Figure I1: CAPM alpha of Series A for each alternative specification, % p.a., with 95% confidence interval

Note: One rule changed at a time; the annual return is shown on the right. The 3-year and 10-year windows cover different dates. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

Table I1: Results for each alternative specification, full Window and Test Period

Specification (one rule changed)	Return	Sharpe ratio	CAPM alpha (t)	Four-factor alpha (t)	Max. drawdown	Test Period return	Test Period CAPM alpha (t)
Base case	**13.41%**	**0.50**	**+1.07 (0.18)**	**+2.26 (0.37)**	**−32.2%**	**22.94%**	**+1.69 (0.29)**
Excluding Tuberville	13.31%	0.59	+1.92 (0.54)	−0.01 (−0.00)	−24.6%	26.21%	+4.71 (0.94)
Excluding Armstrong	13.15%	0.49	+0.84 (0.15)	+2.03 (0.33)	−32.2%	22.38%	+1.18 (0.20)
Excluding Mullin	18.64%	0.64	+5.79 (0.78)	+7.12 (0.96)	−32.2%	35.70%	+10.06 (1.03)
Excluding Boozman	13.55%	0.51	+1.21 (0.21)	+2.40 (0.39)	−32.2%	23.24%	+1.90 (0.31)
Excluding Whitehouse	13.52%	0.51	+1.14 (0.20)	+2.27 (0.37)	−32.2%	22.94%	+1.51 (0.26)
Excluding first 180 days in office	9.43%	0.36	−2.28 (−0.43)	−1.12 (−0.19)	−31.8%	14.80%	−3.24 (−0.86)
Excluding bulk filings (50+ transactions)	11.38%	0.40	−0.98 (−0.14)	+0.80 (0.12)	−32.2%	20.13%	−3.87 (−0.52)
Excluding ETFs	13.41%	0.50	+1.11 (0.18)	+2.18 (0.33)	−32.9%	22.95%	+1.47 (0.24)
Including spouse and child accounts	14.12%	0.59	+1.60 (0.54)	+2.95 (0.95)	−26.7%	23.18%	+2.58 (0.85)
Size at lower bound of band	13.41%	0.48	+1.08 (0.16)	+2.60 (0.35)	−35.7%	23.45%	+1.65 (0.25)
Equal amount per trade	12.42%	0.57	+0.88 (0.39)	+0.72 (0.37)	−23.5%	20.53%	+1.31 (0.43)
Same-day entry before 15:30	13.89%	0.52	+1.48 (0.25)	+2.64 (0.43)	−32.1%	23.22%	+1.90 (0.33)
Equal weight per position	12.50%	0.57	+0.77 (0.32)	+0.73 (0.36)	−22.9%	20.46%	+1.72 (0.62)
3-year window (from Sep 2023)	25.50%	1.11	+2.36 (0.45)	+1.96 (0.31)	−20.1%	35.81%	+10.57 (1.75)
10-year window (from Sep 2016)	15.57%	0.63	−1.23 (−0.30)	+0.15 (0.04)	−51.1%	6.11%	−5.08 (−1.49)
Excluding OTC stocks	13.30%	0.50	+0.97 (0.17)	+2.17 (0.35)	−32.2%	22.71%	+1.47 (0.25)
Amendments applied retroactively	13.20%	0.49	+0.84 (0.14)	+2.23 (0.36)	−32.6%	22.63%	+1.31 (0.22)
Unpriced stocks earn SPY's return	13.42%	0.50	+1.07 (0.19)	+2.25 (0.37)	−32.0%	22.95%	+1.69 (0.29)
All ETFs, incl. bond and T-bill funds	11.97%	0.45	−0.13 (−0.02)	+1.12 (0.18)	−32.2%	19.94%	−0.09 (−0.02)
No warm-up period (a)	15.43%	0.58	+2.99 (0.52)	+4.47 (0.72)	−32.4%	23.23%	+1.70 (0.29)
Yahoo prices only (a)	10.81%	0.41	−1.48 (−0.33)	−0.74 (−0.16)	−32.2%	22.95%	+1.66 (0.28)
Including preferred shares (a)	13.41%	0.50	+1.07 (0.18)	+2.26 (0.37)	−32.2%	22.94%	+1.69 (0.29)
CRSP prices (a)	not run						
Sale on departure from office (b)	15.98%	0.57	+3.30 (0.48)	+4.36 (0.60)	−32.5%	27.83%	+4.07 (0.47)

Note: Alphas in % p.a., Newey-West t. (a) Defined after the Design Period was reviewed. (b) Added after the Test Period was opened. CRSP price data could not be retrieved programmatically under our WRDS access. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.

J. Sources of backtest bias and safeguards

Table J1: Common sources of bias in backtests and how each was addressed

Risk	Safeguard	Effect on Series A
Look-ahead bias	Every input dated to its publication; entry the day after receipt; amendments as first published; benchmark universe formed from filings already public	Same-day entry +0.5 pp; amendments applied retroactively −0.2 pp
Survivorship bias	Scope based on official service dates, so former senators are included; Bloomberg prices for 95% of the missing dollars	Bloomberg prices +2.6 pp; unpriced stocks at most 2.2% of capital
Choice of window	Five-year window fixed in advance; results reported year by year (Table 4)	3 years 25.5%; 10 years 15.6%, alpha −1.2%
Overfitting	Alternatives and tests defined before results; every run reported; adjustment for multiple testing	Range 9.4% to 25.5%; no CAPM alpha with $\|t\| \geq 2$
Leverage mistaken for skill	Risk-matched SPY reported beside every return	−2.9 pp p.a. against risk-matched SPY
Result without economic rationale	Counterparty identified and predictions tested before the Test Period was opened	No evidence of informed purchases (Table 5)

Note: Changes in the annual return of Series A versus the base case; pp = percentage points. Sources: US Senate eFD, Yahoo Finance, Bloomberg, Kenneth French Data Library.